

# Chapter 1

## Groups

We naturally think of many sets as having an associated operation. Consider the familiar set of integers and the operation addition. Adding integers gives another integer. Furthermore it does not matter the order that integers are added. There is a special integer, 0, that does not affect other numbers when added to them. We can appreciate that the set of integers under addition has a desirable structure.

We abstract these properties as *group axioms*, and we will see a highly diverse collection of structures with these same similar properties. The ubiquity of groups in areas both within and outside mathematics makes them a central organizing principle of mathematics. The concept of group developed in the early nineteenth century in the study of polynomial equations.

First we formalize the above mentioned “operation”.

### 1.1 Groups

**Definition 1.** Let  $G$  be a set. A (binary) operation<sup>1</sup> on the set  $G$  is a function  $f: G \times G \rightarrow G$ .

- $f$  is just a function sending two elements to a single element. For example, define  $f$  to be addition.  $f(3, 4) = 3+4 = 7$ .

**Example 1.** Define  $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$  as  $x + 1$  for  $x, y \in \mathbb{R}$ .

$$f(1, 2) = 3 \text{ since } 1 + 2 = 3 \text{ and } f(2, -5) = 2 + (-5) = -3 \text{ since, } 2 - 5 = -3.$$

**Definition 2.** A *group*,  $G$ , be a nonempty set with an associated (binary) operation  $\oplus$ , that combines two elements to form another element with the following properties for all  $a, b, c \in G$ .

1. Closure:  $a \oplus b \in G$ .
  2. Associativity:  $(a \oplus b) \oplus c = a \oplus (b \oplus c)$ .
  3. Identity: There exist an  $0_G \in G$  such that  $0_G \oplus a = a \oplus 0_G = a$ .
  4. Inverse: There exists an  $a^{-1} \in G$  such that  $a \oplus (a^{-1}) = 0_G$ .
- We use,  $\oplus$ , because the operation can be many more things than  $+$ . Sometimes juxtaposition is used, e.g.,  $a \oplus b = ab$ , in which case it should not be confused with multiplication.
  - The a set  $G$  with associated operation  $\oplus$  is often denoted  $(G, \oplus)$ . For example the set of rational numbers under addition might be written,  $(\mathbb{Q}, +)$ .

**Example 2.** The set of rational numbers,  $\mathbb{Q}$ , under addition is a group.

*Proof.*  $\mathbb{Q}$  certainly is nonempty; take  $\frac{1}{2} \in \mathbb{Q}$  for instance. We can write any two rational numbers with a common denominator. So WLOG we can write all rational numbers with the same nonzero integer denominator, say  $d$ .

Closure:  $\frac{x}{d} + \frac{y}{d} = \frac{x+y}{d}$ . Since  $x + y \in \mathbb{Z}$  it follows that  $\frac{x+y}{d} \in \mathbb{Q}$ .

Associativity:  $(\frac{x}{d} + \frac{y}{d}) + \frac{z}{d} = \frac{x+y}{d} + \frac{z}{d} = \frac{x+y+z}{d} = \frac{x}{d} + (\frac{y}{d} + \frac{z}{d})$ .

---

<sup>1</sup>When parts of statements are given in parenthesis, it usually means that future statements will leave that part out. So in the from here on we will simply write “operation” as opposed to writing “binary operation”.

Identity:  $0 = \frac{0}{d}$  and  $\frac{0}{d} + \frac{x}{d} = \frac{x}{d} = \frac{x}{d} + \frac{0}{d}$

Inverse: 0 is the denominator so for any  $\frac{x}{d}$  we need a  $\frac{y}{d}$  such that  $\frac{x}{d} + \frac{y}{d} = 0$ . Solve this equation to find  $y = -x$  so for any rational number  $\frac{x}{d}$ ,  $\frac{-x}{d}$  is the additive inverse.

Thus  $\mathbb{Q}$  is a group. □

**Exercise 1.** Show that the set of integers,  $\mathbb{Z}$ , is a group under addition.

**Exercise 2.** Prove that the set of even integers is a group under addition.

**Exercise 3.** Prove that integers of  $n$  multiples for  $n \in \mathbb{Z}^+$  are a group under addition.

**Example 3.**  $\{1, -1, i, -i\}$  where  $i = \sqrt{-1}$  is a group under multiplication. From the table,

| $\times$ | 1    | -1   | $i$  | $-i$ |
|----------|------|------|------|------|
| 1        | 1    | -1   | $i$  | $-i$ |
| -1       | -1   | 1    | $-i$ | $i$  |
| $i$      | $i$  | $-i$ | -1   | 1    |
| $-i$     | $-i$ | $i$  | 1    | -1   |

we can see that 1 is the identity, every element has an inverse ( $-i \cdot i = 1$  for example), and

the set is closed under multiplication. The last property, associativity, follows by associativity of  $\mathbb{C}$ .

To show that a set is not a group, it is usually easiest to find an example where any one of the properties fails.

**Example 4.**  $S = \{1, -1, i\}$  is not a group under multiplication.

*Proof.*  $-1(i) = -i$  and  $-i \notin S$  so it is not closed. Alternatively,  $i \cdot x = 1 \Rightarrow x = -i$ . So the inverse of  $i$  is  $-i$ ,  $-i \notin S$ . □

**Exercise 4.** Prove that the set of all  $2 \times 2$  matrices with determinant 1 and entries from  $\mathbb{R}$  is a group under matrix multiplication.

**Exercise 5.** Show that the following are *not* groups under the given operation.

1.  $S = \{z : z \in \mathbb{Z} \text{ and } z = \sqrt{5}\}$  under addition:  $(S, +)$
2.  $\mathbb{Q}$ , the set of rationals under multiplication:  $(\mathbb{Q}, \times)$
3. The set of odd integers under addition.
4.  $\mathbb{Z}$  under multiplication:  $(\mathbb{Z}, \times)$

**Exercise 6.** Let  $S = \{a + b\sqrt{5} \mid a, b \text{ are not both zero and } a, b \in \mathbb{Q}\}$ . Prove that  $S$  is a group. (Hint: rationalize the denominator to show that the inverse of  $a + b\sqrt{5}$  is in  $S$ .)

**Exercise 7.** Show that  $\{1, 2, 3\}$  under multiplication modulo 4 is *not* a group but that  $\{1, 2, 3, 4\}$  under multiplication modulo 5 is a group. (Example:  $2 \cdot 3 = 6$  and  $6 = 2 \pmod{4}$  so  $[6] = [2]$  and thus  $2 \cdot 3$  is in  $\{1, 2, 3\}$ ).

It is easy to take the next few theorems for granted. Especially the cancellation law. You probably have been using it for some time without giving it much thought. However we cannot assume such properties until it is proven. Imagine trying to solve a simple linear equation without it.

**Theorem 1. Cancellation Law.** *Right and left cancellation hold in a group, i.e.,  $b \oplus a = c \oplus a \iff b = c$  and  $a \oplus b = a \oplus c \iff b = c$  for any elements  $a, b, c$  in the group.*

*Proof.* Let  $G$  be group with elements  $a, b, c$ . The inverses of the elements,  $a^{-1}, b^{-1}, c^{-1}$ , are also in  $G$  as  $G$  is a group. First we show the case for the right. Suppose that  $b \oplus a = c \oplus a$  then

$$\begin{aligned} (b \oplus a) \oplus a^{-1} &= (c \oplus a) \oplus a^{-1} \\ \Rightarrow b \oplus (a \oplus a^{-1}) &= c \oplus (a \oplus a^{-1}) \text{ by associativity} \\ \Rightarrow b \oplus 0_G &= c \oplus 0_G \text{ since } a \oplus a^{-1} = 0_G \\ b &= c \end{aligned}$$

Similarly we can show the case for the left. That  $a \oplus b = a \oplus c \iff b = c$ . □

**Theorem 2.** *A group has one and only one identity element, i.e., a group's identity element is unique.*

*Proof.* Let  $G$  be a group. Since  $G$  is a group it has an identity element  $0_G$ . Now suppose there is some other  $e \in G$  such that  $e \oplus a = a \oplus e = a$  for any  $a \in G$ . We need to show that  $e = 0_G$ . Since  $G$  is a group  $a$  has an inverse  $a^{-1}$ .

$$\begin{aligned} (0_G \oplus a) \oplus a^{-1} &= (e \oplus a) \oplus a^{-1} \text{ since } 0_G \oplus a = e \oplus a = a \\ 0_G \oplus (a \oplus a^{-1}) &= e \oplus (a \oplus a^{-1}) \text{ by associativity} \\ 0_G \oplus 0_G &= e \oplus 0_G \text{ since } a \oplus a^{-1} = 0_G \\ 0_G &= e \end{aligned}$$

□

**Theorem 3.** *An element in a group has one and only one inverse.*

*Proof.* Let  $0_G$  be the unique identity of a group  $G$ . Suppose there exists an  $a^{-1}, x \in G$  such that  $a \oplus a^{-1} = a \oplus x = 0_G$ . Then

$$\begin{aligned} x &= x \oplus 0_G \\ &= x \oplus (a \oplus a^{-1}) \\ &= (x \oplus a) \oplus a^{-1} \\ &= 0_G \oplus a^{-1} = a^{-1} \end{aligned}$$

So  $x = a^{-1}$  as needed. □

- Note that since the identity of a group is its own inverse, i.e.,  $0_G \oplus 0_G = 0_G$ , theorem 2 can be proved as a corollary of theorem 3.

**Exercise 8.** ★ Addition in  $\mathbb{Z}_n$  is defined as  $[a] + [b] = [a + b]$ . Multiplication in  $\mathbb{Z}_n$  is defined as  $[a] \cdot [b] = [a \cdot b]$ . For example,  $\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$  for convenience we write  $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ .  $2 + 3 = 5$  and  $5 = 1 \pmod{4}$  so  $[5] = [1]$ . Show that  $\mathbb{Z}_2 \times \mathbb{Z}_3$  is a group under modular addition.

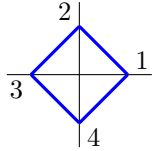
**Definition 3.** A group is *abelian*<sup>2</sup> if  $a \oplus b = b \oplus a$  for all  $a, b$  in the group.

**Example 5.**  $\mathbb{Z}$  is an abelian group under addition.

**Exercise 9.** Prove that the set of all  $n \times n$  matrices is an abelian group under addition.

**Exercise 10.** Prove that the set of all  $2 \times 2$  matrices with determinant 1 and entries from  $\mathbb{R}$  is a non-abelian group under matrix multiplication (we already know that it is a group by exercise 4).

**Exercise 11.** ★ Consider a square placed on the Cartesian plane such that the square's center is at the origin and its corners are on the axes (like a diamond; see the figure below). Any motion of the square which keeps the center at the origin and corners on the axes has the same result as one of eight motions:  $r_0 \equiv$  rotation of  $0^\circ$ ,  $r_1 \equiv$  rotation of  $90^\circ$ ,  $r_3 \equiv$  rotation of  $180^\circ$ ,  $d_x \equiv$   $x$ -axis reflection,  $h_+ \equiv$   $y = x$  reflection, or  $h_- \equiv$   $y = -x$  reflection.



Show that  $D_4 = \{r_0, r_1, r_3, d_x, d_y, h_+, h_-\}$  is a non-abelian group under composition of these motions. The set  $D_4$  is called the *dihedral group of degree 4*.

**Theorem 4.** *Let  $G$  be a group and  $a, b \in G$  then  $(a \oplus b)^{-1} = b^{-1} \oplus a^{-1}$ .*

*Proof.* The inverse of  $(a \oplus b)^{-1}$  is  $(a \oplus b)$ . Now  $(b^{-1} \oplus a^{-1}) \oplus (a \oplus b) = b^{-1} \oplus (a^{-1} \oplus a) \oplus b = b^{-1} \oplus 0_G \oplus b = b^{-1} \oplus b = 0_G$ . So then  $b^{-1} \oplus a^{-1}$  is the inverse of  $(a \oplus b)$ , and by theorem 2 □

**Exercise 12.** Prove that  $G$  is an Abelian group if and only if  $(a \oplus b)^{-1} = a^{-1} \oplus b^{-1}$  for all  $a, b \in G$ .

**Exercise 13.** Suppose that if  $G$  is a group such that if  $a, b, c \in G$  and  $a \oplus b = c \oplus a$  then  $b = c$ . Prove that  $G$  is an Abelian group, i.e., that  $a \oplus b = b \oplus a$  for all  $a, b \in G$ . Hint: consider the element  $a \oplus (b \oplus a) = (a \oplus b) \oplus a$ .

**Exercise 14.** ★ Prove that if in a group  $G$  that  $(a \oplus b)^2 = a^2 \oplus b^2$  then  $G$  is Abelian.

<sup>2</sup>Named in honor of the mathematician Niels Henrik Abel. Abel made many outstanding contributions to a variety of fields; though he died at the age of 26. Abelian groups are also called commutative groups.

## 1.2 Subgroups

**Definition 4.** If a subset  $H$  of a group  $G$  is a group under the operation of  $G$ , then  $H$  is a *subgroup* of  $G$ , denoted  $H \leq G$ .

- Because  $H$  is a group, it is implied in this definition that all subgroups are non-empty.

**Example 6.**  $H = \{2, 4, 6, 8, 0\}$  is a subgroup of the group  $\mathbb{Z}_{10}$  under modular addition.

*Proof.* From example 1, we know that  $\mathbb{Z}_{10}$  is a group under addition modulo 10.  $H \neq \emptyset$  and  $H \subseteq \mathbb{Z}_{10}$ . Using the Cayley table,

| $\oplus$ | 0 | 2 | 4 | 6 | 8 |
|----------|---|---|---|---|---|
| 0        | 0 | 2 | 4 | 6 | 8 |
| 2        | 2 | 4 | 6 | 8 | 0 |
| 4        | 4 | 6 | 8 | 0 | 2 |
| 6        | 6 | 8 | 0 | 2 | 4 |
| 8        | 8 | 0 | 2 | 4 | 6 |

we can see that it is closed, has identity element 0 (the same as  $\mathbb{Z}_{10}$ ), and each element has an inverse ( $6 + 4 \equiv 0 \pmod{10}$  for example). Associativity holds in modular addition; so  $H$  is a group, and hence a subgroup of  $\mathbb{Z}_{10}$ . Written  $H \leq \mathbb{Z}_{10}$ .  $\square$

A subset  $H$  of a group  $G$  will always have the associativity property of  $G$ . So we need only check for closure, existence of an identity and inverses. The relation between the identity and inverses allows us check these properties in a single step with the following theorem.

**Theorem 5. Subgroup Test.** *Let  $G$  be a group and  $H$  be a nonempty subset of  $G$ . Then  $H$  is a subgroup of  $G$  if and only if  $a \oplus b^{-1} \in H$  for all  $a, b \in H$ .*

*Proof.* ( $\Rightarrow$ ) Suppose that  $H \leq G$ , and let  $a, b \in H$ .  $H$  is a group every so element has an inverse, i.e., there exists a  $b^{-1} \in H$ . Groups are also closed so  $a \oplus b^{-1} \in H$ .

( $\Leftarrow$ ) Suppose that  $a \oplus b^{-1} \in H$ , and let  $a, b \in H$  for some nonempty subset of  $G$ .

Associativity: The operation,  $\oplus$ , in  $H$  is the same as that of  $G$  so  $\oplus$  is associative in  $H$ .

Identity: By the hypothesis for any  $a \in H \Rightarrow a \oplus a^{-1} = 0_G \in H$ . So  $H$  has the same identity element as  $G$  (at least one such element exists since  $H \neq \emptyset$ ).

Inverse:  $0_G \in H$  (shown above) so for any  $a \in H$  we have that  $0 \oplus a^{-1} = a^{-1} \in H$ .

Closure: We just showed that for any  $a, b \in H \Rightarrow a^{-1}, b^{-1} \in H$ .  $a \oplus b = a \oplus (b^{-1})^{-1} \in H$ .

So  $H \leq G$ .  $\square$

- Very often, the operation is just addition. Writing  $+$  instead of  $\oplus$ , the subgroup test is then:

A nonempty subset  $H$  of a group  $G$  is a subgroup of  $G$  if and only if  $a - b \in H$  for all  $a, b \in H$ .

**Example 7.**  $H = \{2, 4, 6, 8, 0\}$  is a subgroup of the group  $\mathbb{Z}_{10}$  under modular addition.

*Proof.* For any  $a, b \in H$ ,  $b^{-1} = -b$  since  $b - b = 0$ . Using the subgroup test,  $a - b \in G$  since  $\mathbb{Z}_{10}$  is a group. So we need only show that  $2|a - b$ .  $2|a$  and  $2|b \Rightarrow a = 2n$  and  $b = 2m$  for some  $m, n \in \mathbb{Z}$ , and  $2|2(n - m) \Rightarrow 2|a - b$ . So then  $a - b \in H$ .  $\square$

**Example 8.** Show that  $\mathbb{Q}$  is a subgroup of the group  $\mathbb{R}$  under addition.

*Proof.*  $\mathbb{Q}$  is a nonempty subset of  $\mathbb{R}$ . Let  $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$ .  $\frac{c}{d}^{-1} = -\frac{c}{d}$  since  $\frac{c}{d} - \frac{c}{d} = 0$ .  $\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd} \in \mathbb{Q}$ . So then  $\mathbb{Q} \leq \mathbb{R}$  by the subgroup test.  $\square$

**Exercise 15.** Let  $(\mathbb{Q}, +)$  be the group of rational numbers under addition.

1. Is  $\mathbb{Z}$  a subgroup of  $(\mathbb{Q}, +)$ ?
2. Is  $\mathbb{Z}^+ = \{0, 1, 2, 3, \dots\}$  a subgroup of  $(\mathbb{Q}, +)$ ?

**Exercise 16.** Determine if the following are subgroups.

1. Is  $\mathbb{Q}$  a subgroup of the group,  $(\mathbb{C}, +)$ , the complex numbers under addition?
2. Is  $\mathbb{Z} - \{0\}$  a subgroup of the group,  $(\mathbb{Q} - \{0\}, \cdot)$ , the non-zero rational numbers under multiplication?

**Exercise 17.** Prove that the intersection of two subgroups  $H$  and  $K$  of a group  $G$  is a subgroup.

**Exercise 18.** Let  $H_1, H_2, \dots$  be subgroups of  $G$ . Prove that  $H_1 \cap H_2 \cap \dots$  is a subgroup of  $G$ .

**Exercise 19.** Recall that  $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$  is a group under modular addition.

1. Show that  $H = \{0, 2, 4\}$  and  $K = \{0, 3\}$  are subgroups of  $\mathbb{Z}_6$ .
2. Show that  $H \cup K$  is *not* a subgroup of  $\mathbb{Z}_6$ .
3. Is the union of two subgroups of  $G$  necessarily a subgroup  $G$ ?

**Exercise 20. ★** Let  $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$  and  $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ .  $\mathbb{Z}_6 \times \mathbb{Z}_4$  is a group under modular addition. Show that  $H = \{(0, 0), (3, 0), (0, 2), (3, 2)\}$  is a subgroup.

**Exercise 21.** Let  $G$  be an abelian group and  $H$  be a subgroup of  $G$ . Prove that  $S(H) = \{x | x \in G \text{ and } x \cdot x \in H\}$  is also a subgroup of  $G$ .

**Exercise 22.** Let  $G$  be a group. Prove that  $G$  has a subgroup with one element. This group is called the *trivial group*.

## 1.3 Finite Groups and Cyclic Groups

Groups with finitely many elements have interesting arithmetic properties.

**Definition 5.** The number of elements in a group its *order*, denoted  $|G|$ .

This is exactly how we defined the order of a set. For example,  $|\mathbb{Z}_{10}| = 10$  and from example 6,  $|H| = 5$ .

**Definition 6.** The *order of an element*  $a$  in a group  $G$  is the smallest positive integer  $n$  such that  $a^n = 0_G$  ( $a^n = \underbrace{a \oplus a \oplus \dots \oplus a}_{n \text{ times}}$ ).

**Example 9.** From example 6,  $2 + 2 + 2 + 2 + 2 = 10 \equiv 0 \pmod{10}$ . So in  $H$ ,  $|2| = 5$ . In  $\mathbb{Z}_{10}$ ,  $|2| = 5$  and  $|1| = 10$ .

**Definition 7.** The set *generated by*  $a$  is the set  $\langle a \rangle = \{a^n : n \in \mathbb{Z}\}$ . The element  $a$  is called the *generator* of  $\langle a \rangle$ .

- If  $a$  is an element of a group  $G$  with  $|a| = n$  then  $a^k \in G$  for any integer  $k$ , and  $\langle a \rangle = \{0_G, a, a^2, \dots, a^{n-1}\}$ .

**Example 10.** Let  $2 \in \mathbb{Z}_{10}$ .  $\langle 2 \rangle = \{2, 3, 6, 8, 0\}$ . Recall that we have already showed that this is a subgroup of  $\mathbb{Z}_{10}$ . Since  $\langle 2 \rangle = \{2, 3, 6, 8, 0\}$  we say that 2 generates the group  $\{2, 3, 6, 8, 0\}$ .

**Example 11.** Find all generators of  $\mathbb{Z}_6$ .

$$\langle 1 \rangle = \{1, 2, 3, 4, 5, 0\} \text{ (note: 1 is a generator of } \mathbb{Z}_n \text{ for any } n) \text{ and } \langle 5 \rangle = \{5, 4, 3, 2, 1, 0\} \text{ so } \langle 1 \rangle = \langle 5 \rangle = \mathbb{Z}_6.$$

**Exercise 23.** Find all generators of  $\mathbb{Z}_8$  and  $\mathbb{Z}_{10}$ .

Some of your favorite sets are generated by a single element. For example, any integer  $n$  can be written as  $\pm 1 \pm 1 \pm \dots \pm 1 = n$ . So 1 generates  $\mathbb{Z}$ . Similarly 2 and 3 generate  $2\mathbb{Z}$  and  $3\mathbb{Z}$  respectively.

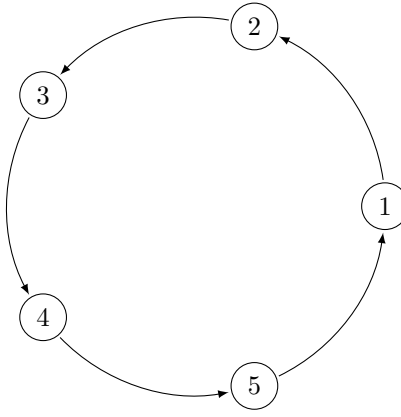
**Definition 8.** Let  $G$  be a group. If there is a single element  $g \in G$  such that every element in  $G$  is equal to  $g^n = \underbrace{g \oplus g \oplus \dots \oplus g}_{n \text{ times}}$  for some  $n \in \mathbb{Z}$  then  $G$  is a *cyclic group* and  $g$  is called the *generator* of  $G$ , and we write  $G = \langle g \rangle$ .

- $\langle g \rangle = \{g^n : n \in \mathbb{Z}\}$

**Exercise 24.** Prove that  $\mathbb{Z}$  is a cyclic group.

**Exercise 25.** Prove that  $\mathbb{Z}_n$  is a cyclic group under modular addition for any  $n \in \mathbb{Z}$  such that  $n > 0$ . (you can use the fact that it is a group).

**Theorem 6.** All cyclic groups are abelian.

Figure 1.1: A graph representing the cyclic group  $\mathbb{Z}_6$ .

*Proof.* Let  $G = \langle a \rangle$  be a group. So any  $x, y \in G$  can be written as  $a^i$  and  $a^j$  respectively.  $a^i \oplus a^j = a^{i+j} = a^{j+i} = a^j \oplus a^i$ .  $\square$

**Theorem 7.** If  $G$  is a group and  $a \in G$  then  $\langle a \rangle \leq G$ .

*Proof.*  $a \in \langle a \rangle$  so  $\langle a \rangle$  is nonempty, and  $G$  is closed so  $a^m \in G$  for any integer  $m$ . So  $\langle a \rangle$  is a nonempty subset of  $G$ . Now we use the subgroup test (theorem 5).

Let  $a^i$  and  $a^j$  be two elements in  $\langle a \rangle$ .  $a^i \oplus a^{-j} = a^{i-j} \in \langle a \rangle$  since  $i - j \in \mathbb{Z}$ .  $\square$

- Theorem 6 gives us another way to show that a group is abelian.
- Theorem 7 gives us an easy way to find abelian subgroups of any group.

**Example 12.**  $\langle 2 \rangle, \langle 3 \rangle, \dots, \langle n \rangle$  are all abelian subgroups of  $\mathbb{Z}$ .

**Theorem 8.** Let  $G$  be a group and  $a \in G$ . If  $|a| = n$  then  $a^i = a^j$  if and only if  $n|i - j$ .

*Proof.* ( $\Leftarrow$ ) Let  $n|i - j$ . Then  $i - j = nq$  and

$$a^{i-j} = a^{nq} = (a^n)^q = (0_G)^q = 0_G$$

as needed.

( $\Rightarrow$ )  $a^i = a^j$  implies that  $a^i \oplus a^{-j} = a^{i-j} = 0_G$ . By the division algorithm we can write  $i - j = qn + r$  for  $q, r \in \mathbb{Z}$  and with  $0 \leq r < n$ .

$$0_G = a^{i-j} = a^{qn+r} = (a^n)^q \oplus a^r = (0_G)^q \oplus a^r = a^r = 0_G$$

Thus  $r = 0$ , and  $i - j = qn \Rightarrow n|i - j$  as needed.  $\square$

- An immediate consequence of this theorem is that the order of the group generated by  $a$  is the same as the order of  $a$ , i.e.,  $|a| = |\langle a \rangle|$ .

**Example 13.** Consider  $\mathbb{Z}_{30}$  a group under addition modulo 30.  $\mathbb{Z}_{30}$  has identity 0 and is generated by 1.  $|5| = 6$  since

$$5 + 5 + 5 + 5 + 5 + 5 = (6)5 \equiv 0 \pmod{30}$$

What is  $5^{40} \pmod{30}$ ?  $5^4 = 4(5) = 20$  and  $6|40 - 4$  so then  $5^{40} \equiv 20 \pmod{30}$  as well. More generally,  $6|x - 4 \iff x = 6q + 4$  for some  $q \in \mathbb{Z}$ . So then  $5^{6q+4} \equiv 20 \pmod{30}$ .

**Corollary 1.** Let  $G$  be a group and  $a \in G$ .  $|a| = n$  and  $a^k = 0_G$  if and only if  $n|k$ .

*Proof.*  $a^k = a^{k-0} = 0_G$  we know by the theorem above that  $n|k - 0$  so  $n|k$ . The other direction also follows from the theorem.  $\square$

## 1.4 Permutation Groups

Permutation groups are functions from a set  $A$  back onto  $A$ .<sup>3</sup> They can be thought of ways of “rearranging” the elements of a set.

**Definition 9.** A *permutation* of a set  $A$  is a function from  $A$  to  $A$  that is one-to-one and onto (a bijection). A *permutation group* of a set  $A$  is a set of permutations of  $A$  that is a group under function composition.

**Example 14.** Let  $\sigma: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$  and  $\gamma: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$  be two bijections from  $\{1, 2, 3\}$  to  $\{1, 2, 3\}$  such that

$$\begin{array}{ll} \sigma(1) = 2 & \gamma(1) = 2 \\ \sigma(2) = 3 & \text{and } \gamma(2) = 1 \\ \sigma(3) = 1 & \gamma(3) = 3 \end{array}$$

Function composition is then well defined, and  $\sigma \circ \gamma: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$  is also a bijective function from  $\{1, 2, 3\}$  to  $\{1, 2, 3\}$ . For example  $\gamma(\sigma(1)) = \gamma(2) = 1$ .

There are different ways to represent permutations on a set  $A$ . As permutations map every element in  $A$  to one and only one element, the mapping of any element forms a cycle  $a \rightarrow b \rightarrow c \rightarrow \cdots \rightarrow a$ . This cycle may not include all the elements in  $A$ . Indeed it may just be  $a \rightarrow a$ , but it will always end where it started. So a map that takes  $a \rightarrow b \rightarrow c \rightarrow \cdots \rightarrow a$  can be written conveniently as  $(a, b, c, \dots, a)$  and  $a \rightarrow a$  as simply  $a$ . Writing permutations in this way is called *cyclic form*.

**Example 15.** Consider the permutations (from the example 14 above)  $\sigma: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$  and  $\gamma: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$  be two bijections from  $\{1, 2, 3\}$  to  $\{1, 2, 3\}$  such that

$$\begin{array}{ll} \sigma(1) = 2 & \gamma(1) = 2 \\ \sigma(2) = 3 & \text{and } \gamma(2) = 1 \\ \sigma(3) = 1 & \gamma(3) = 3 \end{array}$$

Graphs representing  $\sigma$  and  $\gamma$  are shown in figures 1.2 and 1.3 respectively.

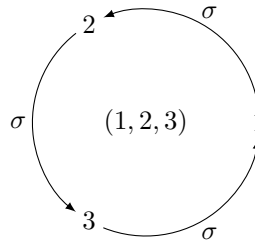


Figure 1.2: A graph representing the permutation of  $\sigma = (1, 2, 3)$  on  $\{1, 2, 3\}$ .



Figure 1.3: Graphs representing the permutation of  $\gamma = (1, 2)(3)$  on  $\{1, 2, 3\}$ .

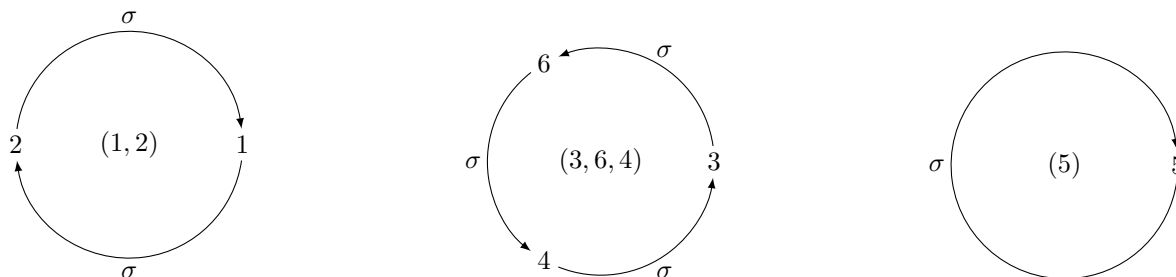
**Example 16.** Let  $\alpha$  be a permutation on  $\{1, 2, 3, 4, 5, 6\}$  defined as follows:

$$\begin{array}{ll} \alpha(1) = 2 & \alpha(4) = 3 \\ \alpha(2) = 1 & \text{and } \alpha(5) = 5 \\ \alpha(3) = 6 & \alpha(6) = 4 \end{array}$$

In cycle form  $\alpha = (1, 2)(3, 6, 4)(5)$ . See the graph in figure

Let  $\alpha$  be a permutation on  $\{1, 2, 3, 4, 5, 6\}$  defined as follows:

<sup>3</sup>Permutations were the first groups studied by mathematicians. In 1832 the French mathematician Évariste Galois gave a criterion for the solvability of some polynomial equations in terms of the symmetry group of its roots (solutions), and the elements of these groups correspond to certain permutations of the roots. Arthur Cayley's work on permutations in 1854 gave the first abstract definition of a finite group.

Figure 1.4: Graphs representing the permutation of  $\alpha = (1, 2)(3, 6, 4)(5)$  on  $\{1, 2, 3, 4, 5, 6\}$ .

$$\begin{array}{ll} \alpha(1) = 5 & \alpha(4) = 6 \\ \alpha(2) = 3 & \text{and} \quad \alpha(5) = 2 \\ \alpha(3) = 1 & \alpha(6) = 4 \end{array}$$

Write in  $\alpha$  in cycle form.

What we label the elements in  $A$  is really arbitrary as a permutation only maps elements to elements, and any meaning the labels might otherwise have is irrelevant to this mapping. For example, both the sets  $\{bob, sue, bill\}$  or  $\{a, b, c\}$  can be represent by the set  $\{1, 2, 3\}$  as they each have three elements. As such we just as well use the set  $\{1, 2, \dots, n-1, n\}$ . Each permutations on a set  $A = \{1, 2, \dots, n-1, n\}$  is an element in the collection of all permutations on  $A$ . A collection we will call the *symmetric group of degree  $n$* , denoted  $S_n$ .

Of course we first need to show that  $S_n$  is a group. We have already seen an example of the group operation we will use, function composition, in example 14. Note that function operation will always be well defined as the permutations are all bijections. This operation is can easily be performed using cycle notation.

**Example 17.** Let  $\sigma = (1, 2, 3)$  and  $\gamma = (1, 2)(3)$ .  $\sigma \circ \gamma$  or just  $\sigma\gamma$  as we will write is then

$$\begin{array}{ll} 1 \rightarrow 2 \rightarrow 3 & \Rightarrow (1, 3) \\ 2 \rightarrow 1 \rightarrow 2 & \Rightarrow (2) \\ 3 \rightarrow 3 \rightarrow 1 & \Rightarrow (3, 1) \end{array}$$

The first and last cycles are equivalent so he have  $\sigma\gamma = (1, 3)(2)$ .

- Note that by writing a permutation in cycle form it can be inferred that both these function are bijections of the set  $\{1, 2, 3\}$ .

**Example 18.** Let  $\alpha = (1, 2)(3)(4, 5)$  and  $\beta = (1, 5, 3)(2, 4)$ .

$$\alpha\beta = (1, 4)(2, 5, 3)$$

**Exercise 26.** Let  $\alpha = (1, 2)(3)(4, 5)$  and  $\beta = (1, 5, 3)(2, 4)$ . Find  $\beta\alpha$ .

$S_n$  is a nonempty set with a well defined operation. Furthermore it is easy to see that this operation is closed. What would the identity element be?

$$e = (1)(2)(3) \dots (n)$$

(1) means that  $1 \rightarrow 1$ . So then  $\sigma e = e\sigma$  for any permutation  $\sigma$  in  $S_n$ .

What about inverses? Permutations are all one-to-one, so each will have an inverse function. Thus  $S_n$  is a group.

**Example 19.** Find the inverse of  $\beta = (1, 5, 3)(2, 4)$ . We need  $1 \rightarrow 5 \rightarrow 1$ . Also we need  $3 \rightarrow 1 \rightarrow 3$ , and  $5 \rightarrow 3 \rightarrow 5$  so  $(5, 1, 3) \in \beta^{-1}$ . Similarly  $2 \rightarrow 4 \rightarrow 2$  and  $4 \rightarrow 2 \rightarrow 4$  implies that  $(2, 4) \in \beta^{-1}$ . Giving,

$$\beta^{-1} = (5, 1, 3)(2, 4) = (1, 3, 5)(2, 4).$$

It is easy to check that  $\beta\beta^{-1} = (1)(2)(3)(4)(5) = \beta^{-1}\beta$ .

**Exercise 27.** Find the inverse of  $\alpha = (1, 2)(3)(4, 5)$  and  $\alpha\beta = (1, 4)(2, 5, 3)$ .

It is easy to find the number of permutations in  $S_n$ . Let  $\sigma \in S_n$  and  $\sigma(x) = y$  represent the permutation mapping  $x \rightarrow y$ . There are  $n$  choices for  $\sigma(1)$ , then  $n-1$  choices for  $\sigma(2)$ , next there are  $n-2$  choices for  $\sigma(3)$  etc. So there will be a total of  $n(n-1)(n-2) \dots 1 = n!$  permutations in  $S_n$ , i.e.,  $|S_n| = n!$ .

**Exercise 28.** Find the order of  $S_4$ ,  $S_1$ , and  $S_3$ .

**Exercise 29.** Show by example that  $S_3$  is not abelian. Thus in general symmetric groups are not abelian.



## 1.5 Cosets

Recall that the elements in  $\mathbb{Z}$  each fell into  $n$  classes of  $\mathbb{Z}_n$  denoted  $[a]$  for an  $a \in \mathbb{Z}_6$ . For  $\mathbb{Z}_6$ ,  $2 \equiv 8 \pmod{6}$  so then  $8 \in [2]$ . We can easily find all the elements in  $[2]$  by repeatedly adding 6 (since we are working with modulo 6). Similarly we could find all the elements of each class.

$$\begin{aligned} [2] &= \{\dots, -2, 0, 2, 8, 14, \dots\} \\ [3] &= \{\dots, -3, 0, 3, 6, 15, \dots\} \\ &\vdots \\ [a] &= \{\dots, a - n, a, a + n, a + 2n, \dots\} \\ &\vdots \\ [6] &= \{\dots, -6, 0, 6, 12, \dots\} \end{aligned}$$

The elements of each class of  $[a]$  can each be expressed as  $a + kn$  for some  $k \in \mathbb{Z}$ . It thus would make sense to write  $a + 6\mathbb{Z}$  instead of  $[a]$ .

We call sets represented like  $3 + 6\mathbb{Z} = \{\dots, -3, 0, 3, 6, 15, \dots\}$  *cosets* (note that  $6\mathbb{Z}$  is a subgroup of  $\mathbb{Z}$ ). The idea was first explored by Galois and is key to proving Lagrange's theorem (the subject of the next section).

**Definition 10.** Let  $H$  be a subgroup of  $G$  and  $a \in G$ .

The set  $a \oplus H = \{a \oplus h : h \in H\}$  is called the *left coset of  $H$  in  $G$  containing  $a$* .

The set  $H \oplus a = \{h \oplus a : h \in H\}$  is called the *right coset of  $H$  in  $G$  containing  $a$* .

**Example 20.** Let  $G = \mathbb{Z}_8$  and  $H = \langle 2 \rangle = \{0, 2, 4, 6\}$ . Then

$$\begin{aligned} 3 \oplus H = 3 \oplus \langle 2 \rangle &= \{3 + 0, 3 + 2, 3 + 4, 3 + 6\} \\ &= \{3, 5, 7, 1\} \end{aligned}$$

also since  $\mathbb{Z}_8$  is abelian,

$$\begin{aligned} H \oplus 3 = \langle 2 \rangle \oplus 3 &= \{0 + 3, 2 + 3, 4 + 3, 6 + 3\} \\ &= \{3, 5, 7, 1\} \end{aligned}$$

The order of  $2 \in \langle 2 \rangle$ , written  $|2|$ , is 4 since  $2 + 2 + 2 + 2 = 2 \cdot 4 = 8 \equiv 0 \pmod{8}$ .  $|\mathbb{Z}_8| = 8$ , and  $2 \mid 8$ , that is  $|2| \mid |\mathbb{Z}_8|$ .

- We will see that the order of a subgroup (or element) always divides the order of its group.

$4, 6, 0 \in \langle 2 \rangle$  and  $4 \oplus \langle 2 \rangle = \{4, 6, 0, 2\} = 2 \oplus \langle 2 \rangle$ ;  $6 \oplus \langle 2 \rangle = \{4, 6, 0, 2\} = 2 \oplus \langle 2 \rangle$ ; and  $0 \oplus \langle 2 \rangle = \langle 2 \rangle = \{4, 6, 0, 2\} = 2 \oplus \langle 2 \rangle$ . From above  $3 \notin \{0, 2, 4, 6\}$  and  $3 \oplus \langle 2 \rangle \neq 2 \oplus \langle 2 \rangle$ .

- We will see that cosets are either mutually distinct or equal. As all elements in  $\mathbb{Z}_8$  are in  $\{0, 2, 4, 6\}$  or  $\{1, 3, 5, 7\}$  these are the only *distinct* cosets. Meaning that any other coset  $a \oplus \langle 2 \rangle$  will be congruent to one of these two cosets.

**Exercise 30.** Find all the distinct cosets of  $\mathbb{Z}_9$ .

**Exercise 31.** Consider the group  $S_3 = \{(1), (1, 2), (1, 3), (2, 3)\}$  and its subgroup  $H = \{(1), (1, 3)\}$ . The distinct left cosets are

$$\begin{aligned} (1)H &= \{(1)(1), (1)(1, 3)\} = H \\ (1, 2)H &= \{(1, 2)(1), (1, 2)(1, 3)\} = \{(1, 2), (1, 3, 2)\} = (1, 3, 2)H \\ (1, 3)H &= \{(1, 3)(1), (1, 3)(1, 3)\} = \{(1, 3), (1, 3)\} = H \\ (2, 3)H &= \{(2, 3)(1), (2, 3)(1, 3)\} = \{(2, 3), (1, 2, 3)\} = (1, 2, 3)H \end{aligned}$$

From these examples/exercises we can see that cosets have some distinct properties.

**Lemma 1.** Let  $H$  be a subgroup of  $G$  and  $a \in H$ . Then  $a \in a \oplus H$  and  $a \in H \oplus a$ .

*Proof.* Let  $a \in H$ . Since  $H$  is a subgroup it has an identity element  $0_G$ , and  $a = a \oplus 0_G \in a \oplus H$  and  $a = 0_G \oplus a \in H \oplus a$ .  $\square$

**Lemma 2.** Let  $H$  be a subgroup of  $G$ . If  $a \in G$  then  $a \oplus H = H$  and  $H \oplus a = H$  if and only if  $a \in H$ .

*Proof.* First we show the result for  $a \oplus H = H$ .

( $\Rightarrow$ ) Suppose that  $a \oplus H = H$ .  $H$  is a group with the same identity element as  $G$ , say  $0$ .  $a \oplus 0 = a \Rightarrow a \in a \oplus H$  as needed.

( $\Leftarrow$ ) Suppose that  $a \in H$  by the lemma above,  $a \in a \oplus H$ . As  $H$  is close,  $a \oplus h \in H$  for any  $h \in H$ . So then  $a \oplus H \subseteq H$ .

Now let  $h \in H$ . Since  $a$  is also in  $H$  it follows that  $a^{-1} \in H$  and thus  $a^{-1} \oplus h \in H$ .  $h = 0 \oplus h = (a \oplus a^{-1}) \oplus h = a \oplus (a^{-1} \oplus h) \in a \oplus H$ . Thus  $H \subseteq a \oplus H$  and it follows that  $H = a \oplus H$ .

The case for  $H \oplus a = H$  follows similarly.  $\square$

- The results below are all true for right hand cosets and left hand cosets. For convenience they are stated for left hand cosets.

**Theorem 9.** Let  $H$  be a subgroup of  $G$ . If  $a, b \in G$  then:

$$a \oplus H = b \oplus H \text{ or}$$

$$(a \oplus H) \cap (b \oplus H) = \emptyset.$$

*Proof.* Suppose that the cosets are not disjoint, i.e.,  $(a \oplus H) \cap (b \oplus H) \neq \emptyset$ . So there is some  $x \in (a \oplus H) \cap (b \oplus H)$ . Thus there exists an  $h_1, h_2 \in H$  such that  $x = a \oplus h_1$  and  $x = b \oplus h_2$ . Thus

$$\begin{aligned} a \oplus h_1 &= b \oplus h_2 \\ a &= (b \oplus h_2) \oplus h_1^{-1} \\ a &= b \oplus (h_2 \oplus h_1^{-1}) \end{aligned}$$

Which implies that  $a \oplus H = (b \oplus (h_2 \oplus h_1^{-1})) \oplus H = b \oplus (h_2 \oplus h_1^{-1}) \oplus H = b \oplus H$ . As needed. The last equality follows since by closure  $h_2 \oplus h_1^{-1} \in H \iff (h_2 \oplus h_1^{-1}) \oplus H = H$  by lemma 2.  $\square$

**Corollary 2.** Let  $H$  be a subgroup of  $G$  and  $a, b \in G$ .  $a \oplus H = b \oplus H$  if and only if  $a^{-1} \oplus b \in H$ .

*Proof.*  $a \oplus H = b \oplus H \iff H = (a^{-1} \oplus b) \oplus H$  which implies that  $a^{-1} \oplus b \in H$  by the theorem above.  $\square$

**Theorem 10.** Let  $H$  be a subgroup of  $G$  and  $a, b \in G$ .  $|a \oplus H| = |b \oplus H|$

*Proof.* One can show that two sets have the same cardinality by finding a one-to-one map between them. Consider the map  $f: a \oplus H \rightarrow b \oplus H$  defined by  $f(a \oplus h) = b \oplus h$ . By the cancellation property, theorem 1,  $f(a \oplus h_1) = f(a \oplus h_2) \Rightarrow b \oplus h_1 = b \oplus h_2 \Rightarrow h_1 = h_2$  so  $f$  is one-to-one as needed.  $\square$

**Corollary 3.** Let  $H$  be a subgroup of  $G$  and  $a \in G$ .  $|a \oplus H| = |H|$

*Proof.* This follows immediately from theorem 10 above, since  $0_G \in G$ . So then  $|a \oplus H| = |0_G \oplus H| = |H|$ .  $\square$

**Theorem 11.** Let  $H$  be a subgroup of  $G$  and  $a \in G$ .  $a \oplus H$  is a subgroup of  $G$  if and only if  $a \in H$ .

*Proof.* ( $\Rightarrow$ ) Let  $a \oplus H \leq G$ . So then  $0_G \in a \oplus H$ .  $0_G$  is also in  $0_G \oplus H = H$  thus  $(a \oplus H) \cap H \neq \emptyset$  and it follows that  $a \oplus H = H$  by theorem 9.

( $\Leftarrow$ ) Suppose that  $a \in H$ . This implies that  $a \oplus H = H$  by lemma 2, and  $a \oplus H = H \leq G$ .  $\square$

## 1.6 Lagrange's Theorem

Our study of group theory ends with Lagrange's theorem and its applications. Lagrange's theorem is the most important result in finite group theory. It provides a necessary characterization for the order of subgroups (and consequently elements) of a finite group. Recall example 20:

**Example 21.**  $\mathbb{Z}_8$  and  $\langle 2 \rangle = \{0, 2, 4, 6\}$ .

- $\langle 2 \rangle \leq \mathbb{Z}_8$ : Since  $2 \in \mathbb{Z}_8$ ,  $\langle 2 \rangle$  is the cyclic subgroup generated by 2. Recall that it is also abelian as all cyclic groups are abelian (theorem 6).
- $|2| = |\langle 2 \rangle| = 4$ : Since  $2^4 = 2 + 2 + 2 + 2 = 8 \equiv 0 \pmod{8}$ .
- Note that  $4 \mid 8$ : This is no coincidence. We will see that Lagrange's theorem tell us that this is a necessary condition of all subgroups and group elements.

- Recall that cosets of  $\langle 2 \rangle$  are either distinct or disjoint, e.g.,  $g_1 + \langle 2 \rangle = g_2 + \langle 2 \rangle$  or  $(g_1 + \langle 2 \rangle) \cap (g_2 + \langle 2 \rangle) = \emptyset$ . So the distinct left/right cosets of a subgroup *partition* its group. This will be important in the proof of Lagrange's theorem.

**Lemma 3.** *If  $H$  is a subgroup of a finite group  $G$  then  $G$  is the union of the distinct cosets of  $H$ , i.e.,*

$$G = a_1 \oplus H \cup a_2 \oplus H \cup \cdots \cup a_k \oplus H$$

for a list of elements  $a_1, a_2, \dots, a_k \in G$  where  $a_i \oplus H \neq a_j \oplus H$  if  $i \neq j$  for any  $1 \leq i, j \leq k$ .

*Proof.* Let  $a_1$  be any element in  $G$ . If there is no such element  $a_2 \in G$  such that  $a_1 \oplus H \neq a_2 \oplus H$  then  $a_2 \in a_1 \oplus H$  for any  $a_2 \in G$ . Which implies that  $a_1 \oplus H = G$ , and we are done.

Otherwise we can either choose another  $a_2$  such that  $a_1 \oplus H \neq a_2 \oplus H$ . As above, either we can find another element  $a_3 \in G$  such that  $a_3 \oplus H$  will be distinct from  $a_1 \oplus H$  and  $a_2 \oplus H$  or  $a_1 \oplus H \cup a_2 \oplus H = G$ .  $G$  is finite, so continuing like this we can compile a list of distinct elements in  $G$ ,  $a_1, a_2, \dots, a_k$  for  $k \leq |G|$ , such that  $G = a_1 \oplus H \cup a_2 \oplus H \cup \cdots \cup a_k \oplus H$ .  $\square$

While the cosets themselves will be unique the list of elements  $a_1, a_2, \dots, a_k$  is not. However the number of cosets  $k$  will always be the same. This number is important enough to have a name -the *index* of the subgroup  $H$ .

**Definition 11.** If  $H$  is a subgroup of a finite group  $G$  such that  $G = a_1 \oplus H \cup a_2 \oplus H \cup \cdots \cup a_k \oplus H$  then the  $k$  is called the *index* of  $H$  in  $G$ ; denoted  $|G : H|$ . As we will see below,  $|G : H| = |G|/|H|$  so the index is sometimes notated  $|G|/|H|$ .

**Theorem 12. Lagrange's Theorem.** *If  $H$  is a subgroup of a finite group  $G$  then  $|H| \mid |G|$ .*

*Proof.* Let  $a_1 \oplus H, a_2 \oplus H, \dots, a_k \oplus H$  be the distinct left cosets of  $H$  in  $G$ . By lemma 3 above,

$$G = a_1 \oplus H \cup a_2 \oplus H \cup \cdots \cup a_k \oplus H$$

. Since this union is disjoint we have,

$$|G| = |a_1 \oplus H| + |a_2 \oplus H| + \cdots + |a_k \oplus H|.$$

By the corollary to theorem 10, each coset  $a_i \oplus H$  in this union has exactly  $|H|$  elements, i.e.,  $|a_1 \oplus H| = |H|$ . So then  $|G| = k \cdot |H|$ .  $\square$

**Example 22.** Consider  $\mathbb{Z}_8$  and its subgroup  $\langle 2 \rangle$ . The order of  $\langle 2 \rangle$  is either 1, 2, 4, or 8 as these are the only divisors of  $|\mathbb{Z}_8| = 8$ .

**Example 23.**  $H = \{0, 1, 2\}$  is not a subgroup of  $\mathbb{Z}_8$  as  $|H| = 3$  and  $3 \nmid 8$ .

**Example 24.** Let  $G$  be a group and  $H \leq G$ . If  $|G| = 35 = 7 \cdot 5$ . The order of  $H$  is 1, 5, 7, or 35.

**Example 25.** Let  $G$  be a group and  $H \leq G$ . If  $|G| = 40$  and  $|H| = 7$ . Then  $H \not\leq G$ .

Note: The converse of Lagrange's theorem is false. Simply because the order of a subset divides the order of a group does not make it a subgroup.

**Example 26.** (The converse of Lagrange's theorem is false). Let  $H = \{1, 2, 3, 4\} \subseteq \mathbb{Z}_8$ .  $|H| = 4$  and  $4 \mid 8$ , but  $H \not\leq G$  as  $0 \notin H$ . So it has no identity element.

**Corollary 4.** *If  $G$  is a group and  $a \in G$  then  $|a| \mid |G|$ .*

*Proof.* Recall that  $|\langle a \rangle| = |a|$  for any element  $a$  in a group  $G$ , and that  $\langle a \rangle \leq G$ . So as an immediate consequence of Lagrange's theorem  $|a| \mid |G|$ .  $\square$

**Corollary 5.** *If  $G$  is a group with prime order then  $G$  is cyclic.*

*Proof.* As  $|G| > 1$ , we can choose an element  $a \in G$  such that  $a \neq 0_G$ . Since  $|G| = p$  for some prime integer  $p$ ,  $|a| \mid p$  by Lagrange's theorem. Thus  $|a| = 1$  or  $p$ , but  $|a| \neq 1$  since  $a \neq 0_G$ . It follows that  $|\langle a \rangle| = p$ , and since  $\langle a \rangle \subseteq G$  the result follows.  $\square$